



ITI Cmos Analog IC Design

Lab 04

Common Drain Amplifier

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Section 7 online

Part 1: Device Sizing

1.1 Sizing Assistant (ADT)

Design Parameters

Parameter	Value
Input transistor	PMOS
L	1 μ m
V*	200 mV
Quiescent (DC) input voltage	0 V
Supply	1.8 V
Current consumption	10 μ A

1.2 Sizing Result

	Name	TT-27.0
1	ID	10u
2	IG	N/A
3	L	1u
4	W	19.52u
5	VDB	942.8m
6	VGB	942.8m
7	VGS	942.8m
8	VDS	942.8m

	Name	TT-27.0
10	VGB	942.8m
11	VDB	942.8m
12	gm/ID	9.919
13	Vstar	201.6m
14	fT	234.9MEG
15	gm/gds	424.7
16	VA	42.82
17	ID/W	512.3m

	Name	TT-27.0
16	VA	42.82
17	ID/W	512.3m
18	gm/W	5.081
19	AREA	19.52p
20	gm	99.19u
21	gmb	46.92u
22	gds	233.5n
23	fT	234.9MEG

LUT

pmos_03v3

?

Corner

TT

☐ All

?

Temp (°C)

27.0

☐ All

?

Frequency

1

?

ID

10u

?

gm/ID

10

?

L

1u

?

VDS

VGS

?

VSb

0

?

Stack

1

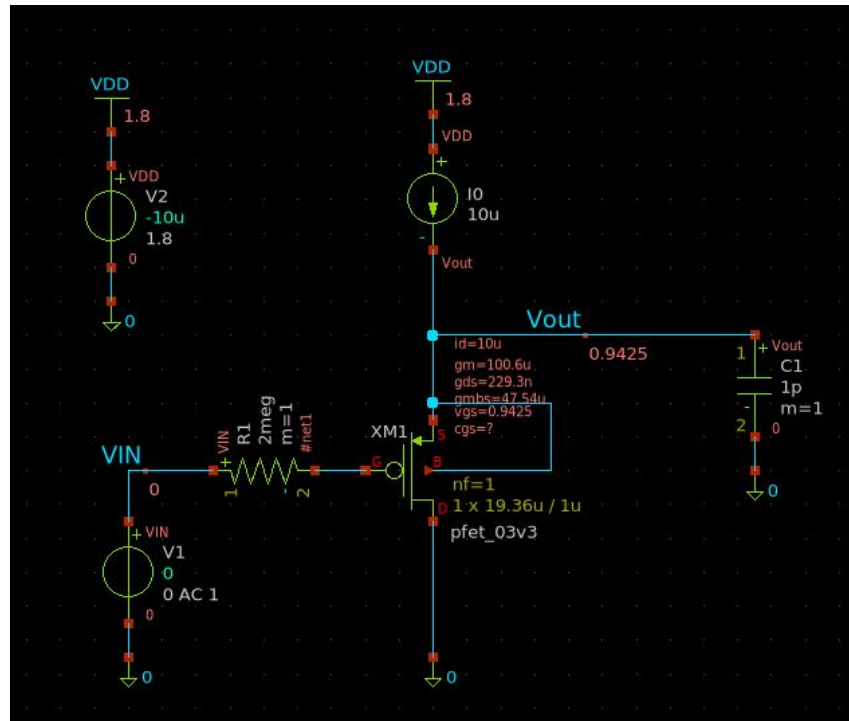
?

ID	L	W	gm	gds
10u	1u	19.52u	99.19u	233.5n

Part 2: CD Amplifier

2.1 OP (Operating Point) Analysis

Schematic



OP Simulation Snapshot

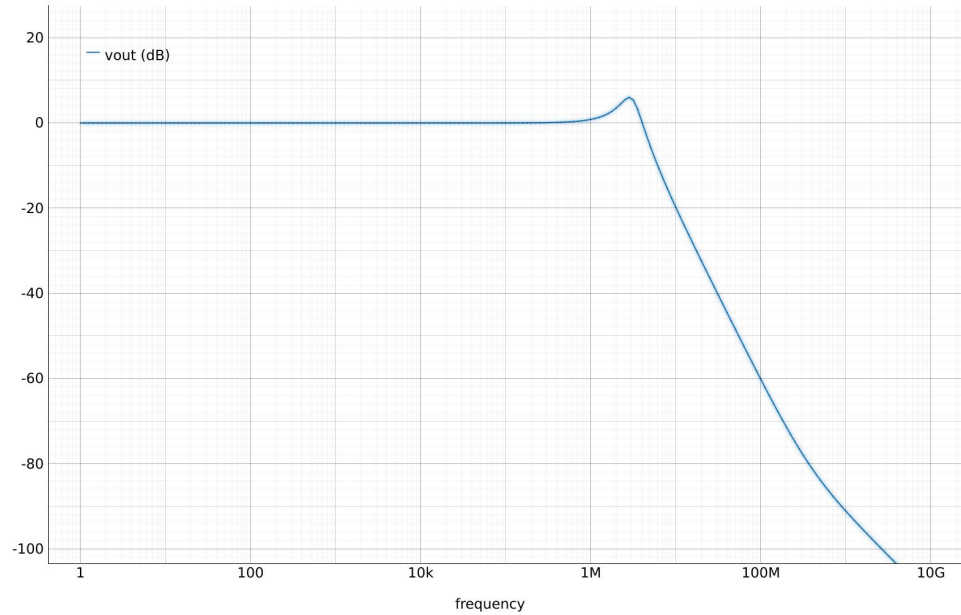
```
device      m.xm1.m0
model       pfet_03v3.13
id          1e-05
gm          0.000100602
gds         2.29268e-07
gmbs        4.7537e-05
vgs         0.942532
vth         0.783939
vds         0.942532
cgs         -4.95661e-14
cgd         -1.38845e-17
```

Saturation Check

All transistors operate in saturation.

2.2 AC Analysis

Bode Plot — Magnitude



There are peaking in frequency domain equal to 6 db

Hand Analysis — Quality Factor

Use the op from op analysis

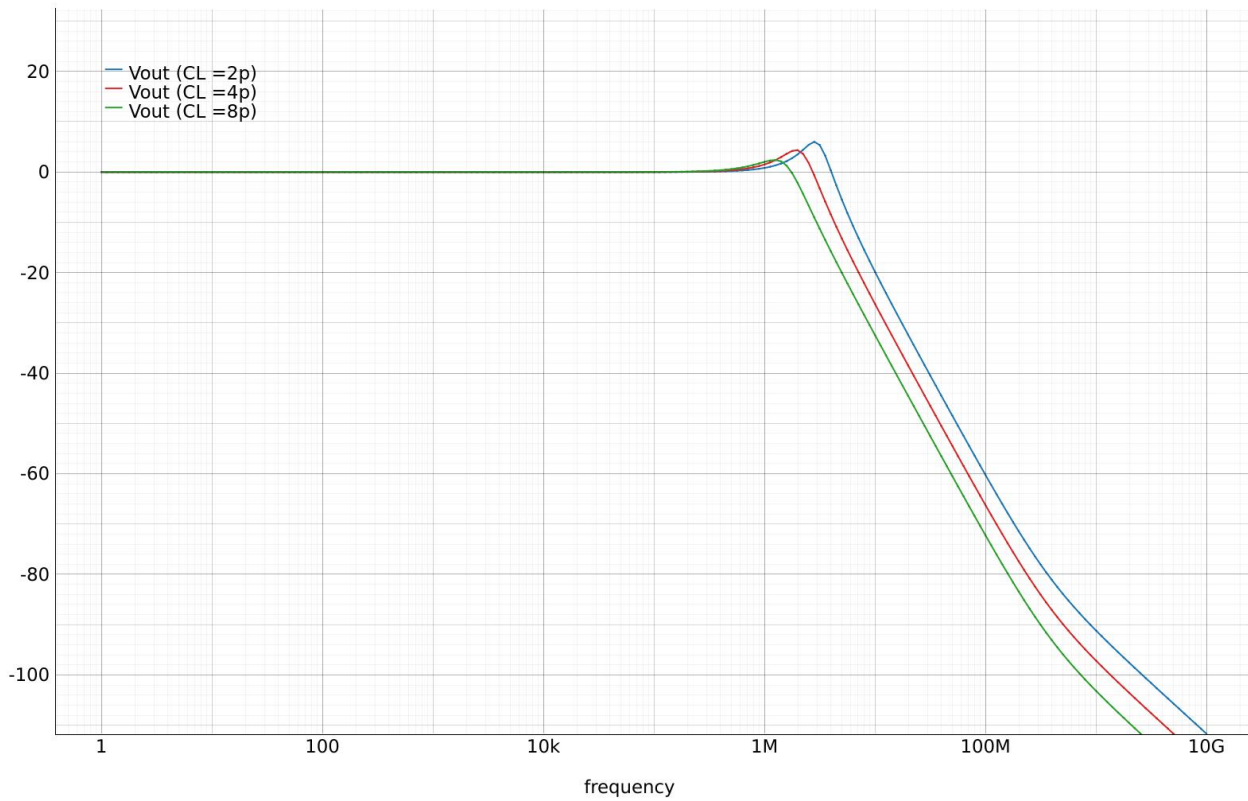
$$Q = \sqrt{\frac{g_m(C_{gs} + C_{gd})R_{sig}}{C_L}} = 2.22$$

$$\text{So } Q > 0.707$$

system is under damped, peaking in frequency domain with ringing in time domain

Parametric Sweep: CL = 2p, 4p, 8p

Bode Plot Magnitude (Overlaid)



Peaking vs CL

CL	Peaking (dB)
2p	5.9966
4p	4.33344
6p	2.3851

Comment

From Q formula $Q = \sqrt{\frac{g_m(C_{gs} + C_{gd})R_{sig}}{CL}}$

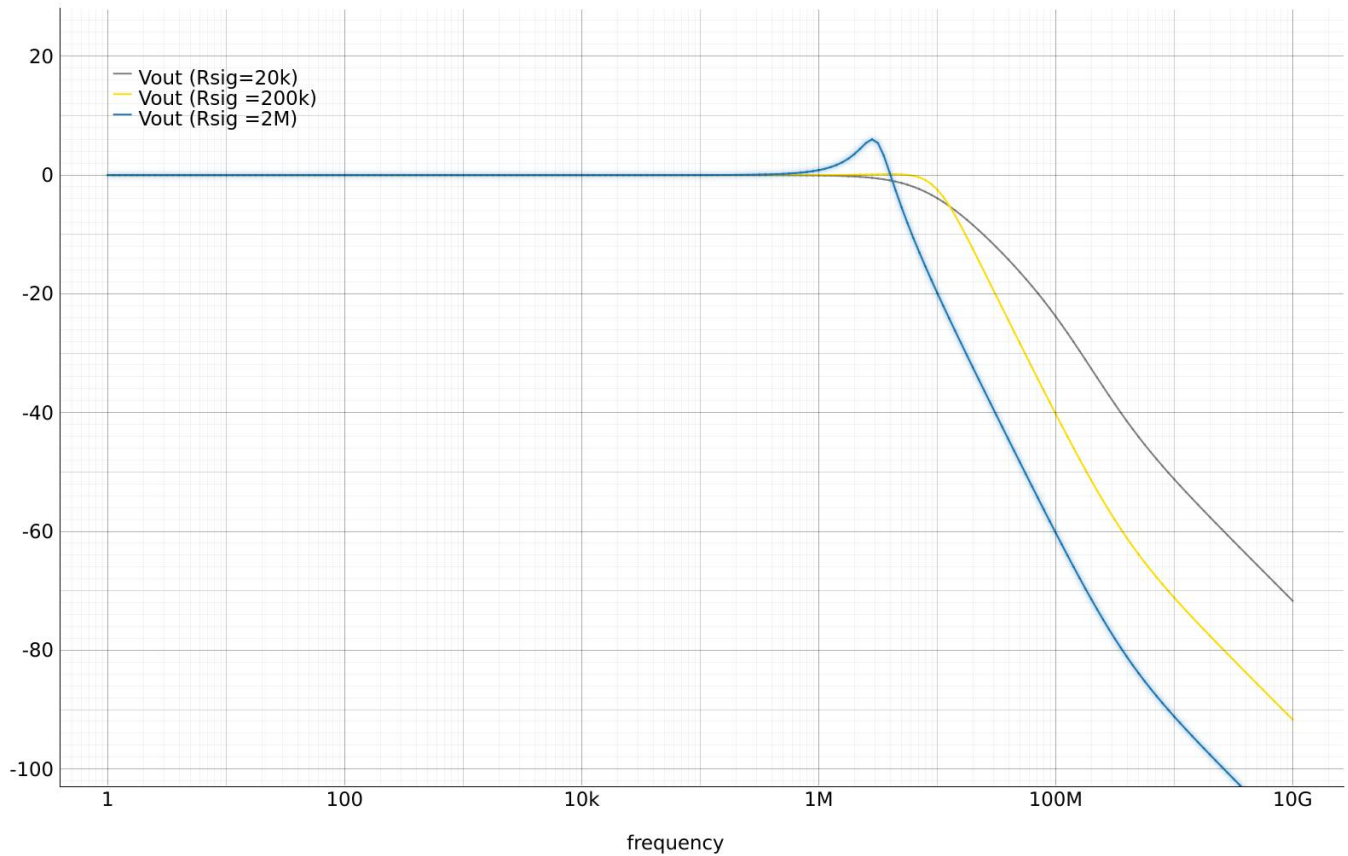
Increasing CL eventually decreases Q that will decrease the peaking due to the dominant pole is moving to a lower frequency .

Also from $\omega_o = \frac{1}{\sqrt{b_2}} \approx \sqrt{\frac{g_m}{(C_{gs} + C_{gd})C_L R_{sig}}}$

increasing CL will decrease dominant pole so bandwidth decrease also .

Parametric Sweep: R_{sig} = 20k, 200k, 2M

Bode Plot Magnitude (Overlaid)



Peaking vs R_{sig}

Rsig	Peaking (dB)
20k	-0.0197724
200k	0.121317
2meg	5.9966

Comment

From Q formula $Q = \sqrt{\frac{g_m(C_{gs} + C_{gd})R_{sig}}{C_L}}$

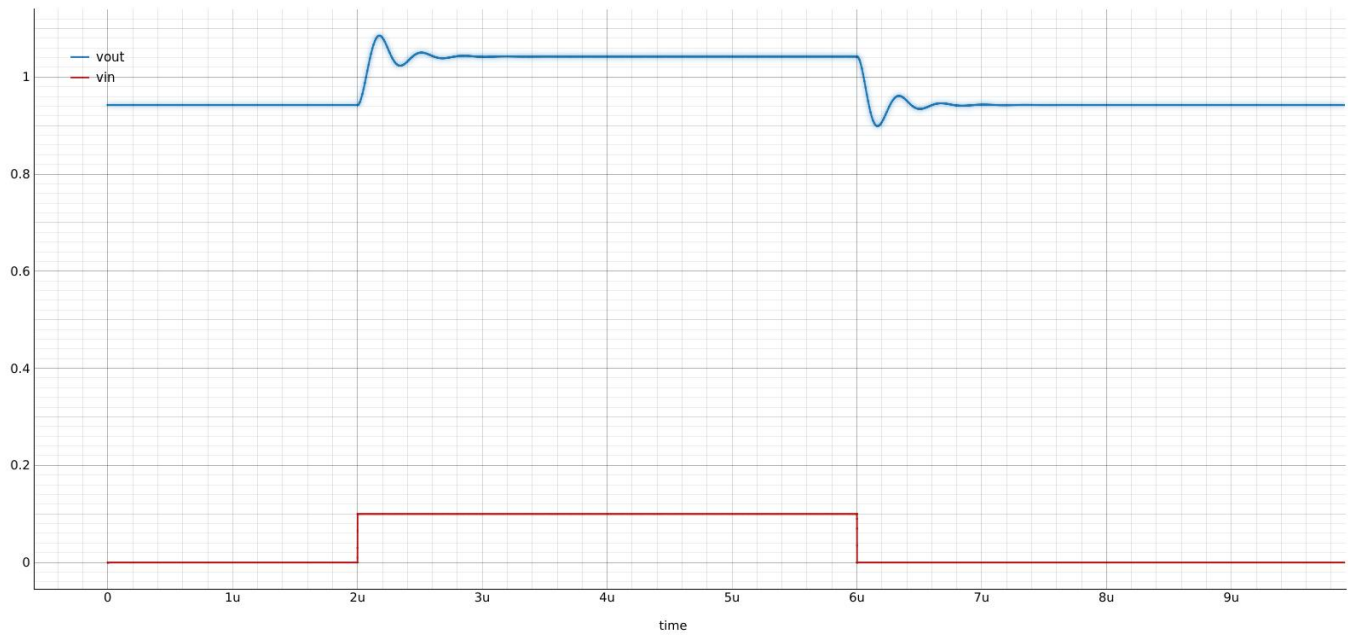
Decreasing R_{sig} will decrease Q that decreasing the peaking in frequency domain

Also from $\omega_o = \frac{1}{\sqrt{b_2}} \approx \sqrt{\frac{g_m}{(C_{gs} + C_{gd})C_L R_{sig}}}$

Decreasing R_{sig} will increase dominant pole so bandwidth increase also .

2.3 Transient Analysis

VSIG and VOUT vs Time



DC Shift Calculation

Vout - Vin = 0.94 V that is equal to VSG (op analysis)

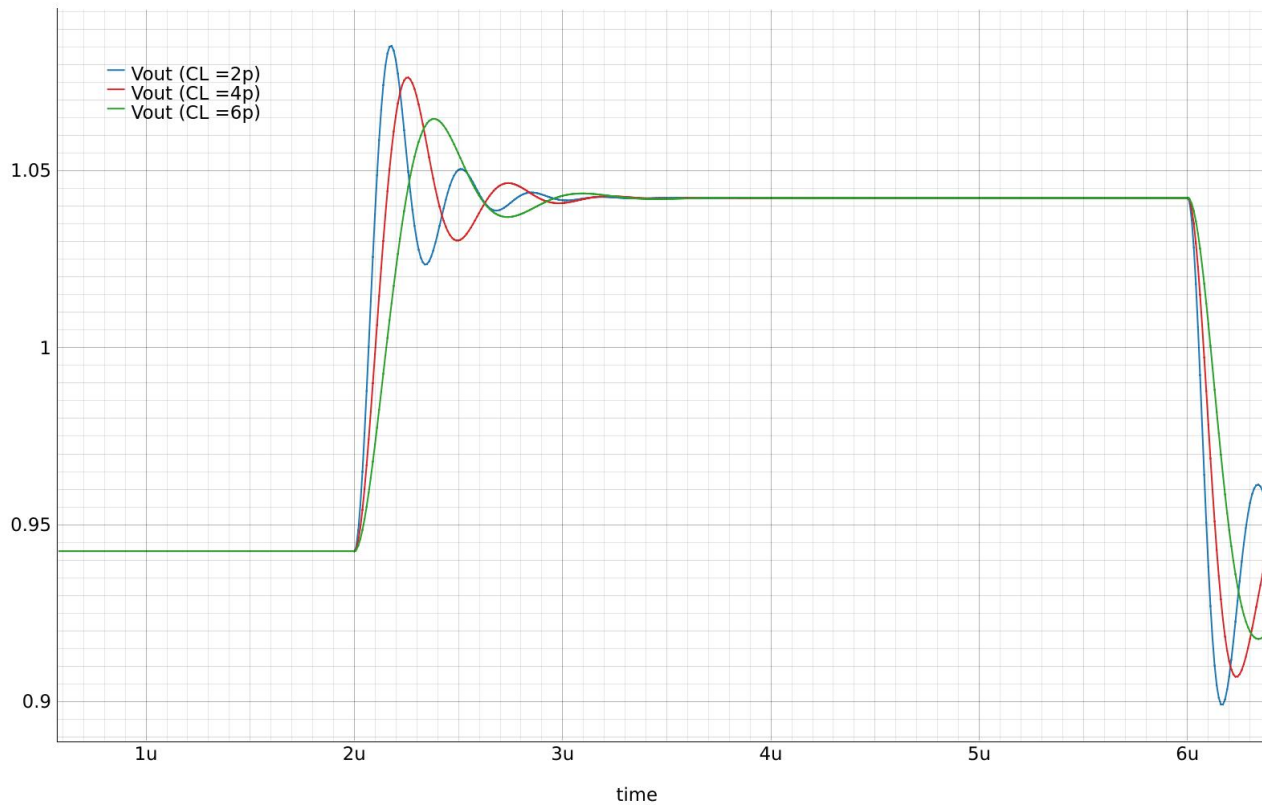
to Shift the output down we should use NMOS , Since VS will be less than VG

The output shows underdamped ringing that is because Q is greater than 0.5

$$Vov = (1.08 - 1.04) / .1 * 100 = 40\%$$

Parametric Sweep: CL = 2p, 4p, 8p

VOUT vs Time (Overlaid)



Overshoot vs CL

CL	Overshoot (%)
2p	43.0583
4p	34.1147
8p	22.4081

Comment

From Q formula $Q = \sqrt{\frac{g_m(C_{gs} + C_{gd})R_{sig}}{CL}}$

Increasing CL will decrease Q that decreases the ringing in time domain

So overshoot will decrease

Parametric Sweep: R_{sig} = 20k, 200k, 2M

V_{OUT} vs Time (Overlaid)



Overshoot vs R_{sig}

R _{sig}	Overshoot (%)
20k	0
200k	6.50424
2meg	43.0583

Comment

From Q formula $Q = \sqrt{\frac{g_m(C_{gs} + C_{gd})R_{sig}}{CL}}$

decreasing R_{sig} will decrease Q that will decrease the ringing in time domain

So overshoot will decrease

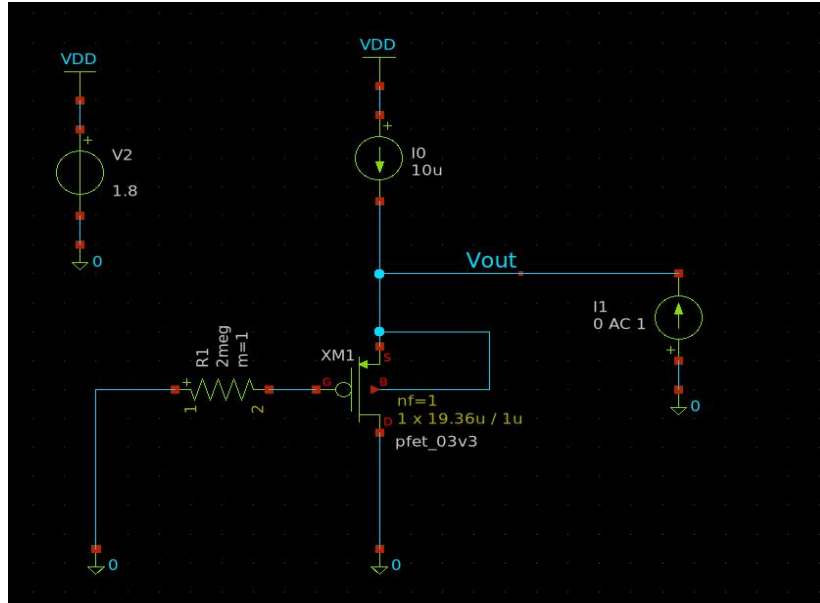
note at R_{sig} = 20k

Q = 0.22 that is less than 0.5, it is over damped so no ringing in time domain

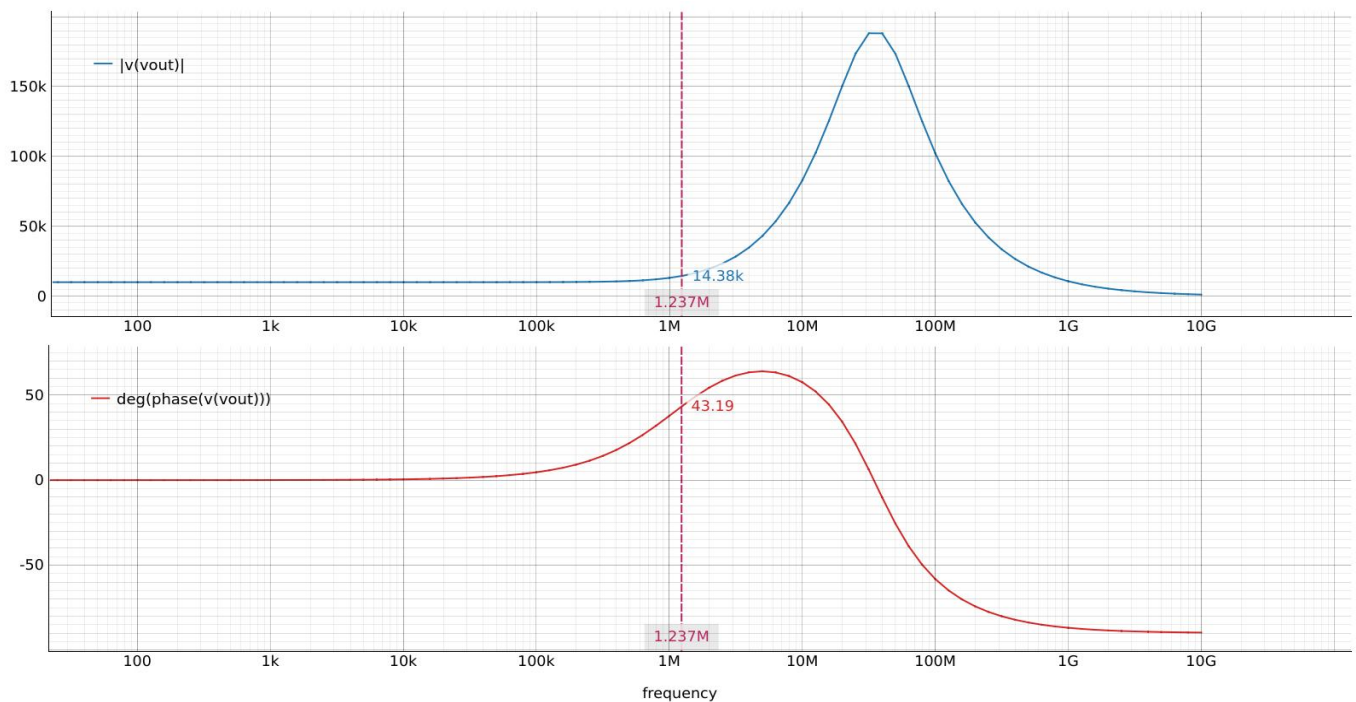
And overshoot is zero.

2.4 Zout (Inductive Rise)

Test Schematic



Zout Magnitude and Phase vs Frequency



Inductive Rise — Why?

There are inductive rise due to R_{sig} is greater than $1/g_m$, there is no physical inductor.

Does Zout Fall at High Frequency? Why?

At high frequency Cgd appears and shunt Rsig ($\approx 1/R_{sig}C_{gd}$?), so zout will fall

Hand Analysis — Zeros, Poles, and Magnitude

Magnitude at low frequency = $1 / g_m = 10k$

Magnitude at medium frequency = $\frac{c_{gs} + c_{gd}}{g_m * c_{gd}} \parallel r_o = 182 k$

$$\omega_z = \frac{1}{2 * \pi * R_{sig} c_{gs}} = 1 MHz$$

$$\omega_p = 30 MHz \quad // \text{ from solving the transfer function}$$

Analytical vs. Simulation Comparison

Quantity	Analytical	Simulated
Mag (low frequency)	10 k	10 k
Mag (high frequency)	182 k	180 k
ω_z	1 MHz	1.23 MHz
ω_p	30 MHz	32 MHz

In high frequency zout must be Rsig but cgd take over and zout decreased as shown .

Analytical result is close to simulated result with error is less than 15%

2.5 [Bonus] Solving the Peaking / Ringing Problem

To solve the problem of peaking, another way used instead of varying the load capacitor or the signal resistance, by the way varying them not practical because they determined by the previous and next stage.

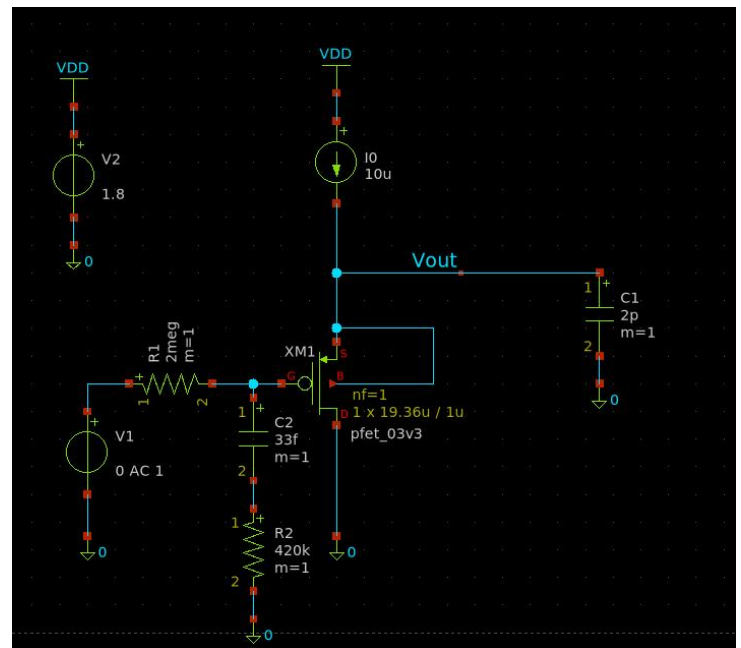
A compensation network is a perfect alternative to solve this problem. Consists of a capacitor and resistor in series put in input node to reshape the transfer function and reduce the Q below the 0.5.

$$C = \frac{gm * cgs * cL}{(gm + 1/ro)(cgs + cl)}$$

$$= 33fF$$

$$R = \frac{(cgs * cL)^2}{(cgs * cl * gm)}$$

$$= 420 K\Omega$$



● *VOUT vs Time*

